
Quantum Field Theory I

Exercise sheet 2

This exercise sheet will be discussed on Oct 19.
Please hand in your solution of the exercise marked with (★)
in the mail box, before the lecture on Monday.

1. The Schrödinger Lagrangian (★)

Consider the Schrödinger Lagrangian

$$\mathcal{L} = i\frac{\hbar}{2}(\Psi^* \partial_t \Psi - \Psi \partial_t \Psi^*) + \frac{\hbar^2}{2m} \nabla \Psi^* \cdot \nabla \Psi - V(\vec{r}) \Psi^* \Psi,$$

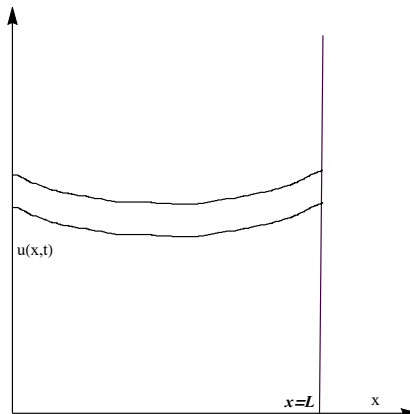
where $\Psi = \Psi(\vec{r}, t)$ is a complex field. Derive the corresponding equation of motion. You can treat Ψ and Ψ^* as independent field components.

2. Lagrangian Principle in the Continuum Limit (★)

As a model for the bending of a beam in one dimension (coordinate x), we consider the following Lagrangian

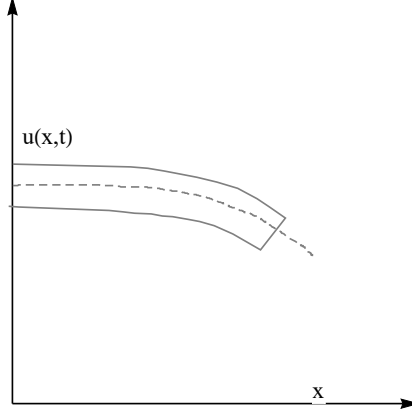
$$\mathcal{L}[u, \dot{u}, u''] = \frac{1}{2} \dot{u}^2 - \frac{1}{2} (u'')^2 + u f(x, t),$$

where $f(x, t)$ is a given function and the x -interval extends from 0 to L . One is looking for the solution $u = u(x, t)$ starting from the variational principle $\delta \mathcal{L} = 0$.



- Give a physical interpretation for the three terms of $\mathcal{L}$.
- Due to the second derivative appearing in the Lagrangian the Euler-Lagrange equations derived in the lecture cannot be applied here. Starting from the action principle, derive the Euler-Lagrange equations for a Lagrangian $\mathcal{L}[u, \dot{u}, u'']$
- Which equation of motion does $u(x, t)$ satisfy?

- d) For a time-independent external force $f = f(x)$, we look for the time-independent solution $u = u(x)$. What is the physical Interpretation of $\delta S = 0$? Find the differential equation that determines $u(x)$.
- e) Consider a beam with a fixed left end, $u(x=0) = u'(x=0) = 0$, bent by the gravitational force, $f(x) = -g$. Find the solution $u(x)$. What are the boundary conditions at $x = L$?



- f) What is the Hamiltonian density of the system?

3. Functional Derivative

Given a functional

$$F : \mathbb{C}^\infty \ni f \mapsto F[f] \in \mathbb{R},$$

i.e. a map from the space of infinitely many times continuously differentiable functions into the real numbers, then the functional derivative is defined as

$$\frac{\delta F[f(x)]}{\delta f(y)} = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F[f(x) + \epsilon \delta(x-y)] - F[f(x)]),$$

from which we can derive the expansion

$$F[f(x) + \epsilon \phi(x)] = F[f(x)] + \epsilon \int dx' \frac{\delta F[f(x')]}{\delta f(x')} \phi(x') + O(\epsilon^2). \quad (1)$$

Relying on the definition and the expansion demonstrate that the following relationship is satisfied:

a)

$$\frac{\delta G[F[f]]}{\delta f(x)} = \int \frac{\delta G[F]}{\delta F(y)} \frac{\delta F[f(y)]}{\delta f(x)} dy \quad (2)$$

Compute $\frac{\delta F[f]}{\delta f(z)}$ for the following functionals $F[f]$:

b) $F[f] = \int dx f(x)$

c) $F[f] = \int dx f(x)^2$

d) $F_x[f] = \int dy G(x,y) f(y)$

$$\mathbf{e)} \quad F[f] = \int \mathrm{d}x \, (\partial_x f(x))^2$$